IVF / Fertility Clinic Management System

Lab Technician

Requirements Specification

Software Engineering – Laboratory Deliverable

Role: Silvja Fejzulla & Arbri Hamzallari (Lab Technician)

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1. Lab Technician Role Overview

The Lab Technician (Silvja Fejzulla & Arbri Hamzallari) is a core clinical role in the IVF / Fertility Clinic Management System. Lab Technicians are responsible for all laboratory-side operations: processing patient samples, conducting and reporting tests, recording embryo development, maintaining the donation bank, and ensuring equipment and quality-control compliance. Their work directly informs clinical decisions made by doctors and determines treatment outcomes for patients.

Core Responsibilities:

- Sample Processing: Receive, label, and track patient and donor samples from arrival through to analysis, maintaining full chain-of-custody at every stage.
- Test Reporting: Process lab test orders requested by doctors, upload completed test reports linked to the patient record, and flag any abnormal results automatically for urgent clinical review.
- Embryology Support: Record daily embryo development observations throughout the IVF culture period, document cryopreservation details for frozen embryo transfer planning, and review doctor instructions for each embryo.
- Donation Bank Management: Maintain accurate sperm, egg, and embryo donation stock levels, update donor sample screening statuses, and ensure expired or rejected samples are removed from the assignable pool.
- Equipment & Quality Control: Keep lab equipment calibration and maintenance records up to date, perform and document QC checks on instruments and reagents, and escalate any compliance failures before patient testing resumes.
- Audit & Compliance: Log all custody events and record changes in a timestamped, immutable audit trail to satisfy regulatory inspection requirements.

Attribute	Detail
Role	Lab Technician
Team Members	Silvja Fejzulla, Arbri Hamzallari
System Modules Accessed	Test Reports, Embryology, Donation Bank, Equipment Management, Sample Tracking, Quality Control
Interacts With	Doctor (receives test requests, shares results), Hospital Administrator (low-stock alerts), Patient (indirect via test results)
Key Constraints	Can only access records for assigned patients/cycles; all entries are timestamped and immutable once confirmed; must authenticate for all inventory changes; must comply with clinical and regulatory traceability requirements.

Access Level: The lab technician has read/write access to test reports, embryology records, donation bank inventory, sample tracking, and equipment management. The lab technician does not have access to edit patient personal data (administrator's domain), appointment schedules, billing, or clinical treatment plans (doctor's domain). Access is restricted to assigned patients and cycles only.

2. Functional Requirements

The following table lists all functional requirements specific to the Lab Technician role. Each requirement is traceable to one or more use cases.

Req #	Requirement	Pri.	Comments / Traceability
FR-LT-01	The system shall display a prioritised queue of pending lab test orders sorted by urgency level and due time.	1	Supports UC_LT_01.
FR-LT-02	The system shall allow the lab technician to receive and label patient samples via barcode scan or manual sample ID entry.	1	Supports UC_LT_01.
FR-LT-03	The system shall allow upload of test result files (PDF, JPEG, PNG, CSV; max 25 MB) linked to the patient record and the originating doctor order.	1	Supports UC_LT_01.
FR-LT-04	The system shall automatically compare uploaded result values against predefined clinical reference ranges and flag any abnormal values with a colour-coded indicator.	1	Supports UC_LT_01.
FR-LT-05	The system shall send an urgent alert to the requesting doctor whenever an abnormal result is flagged.	1	Supports UC_LT_01. Alert within 30 seconds.
FR-LT-06	The system shall notify the requesting doctor when a test report is uploaded and ready for review.	1	Supports UC_LT_01.
FR-LT-07	The system shall allow the lab technician to record daily embryo development observations including development day, cell count, fragmentation percentage, morphology grade, and free-text notes.	1	Supports UC_LT_02.
FR-LT-08	The system shall allow optional attachment of a microscopy image (max 10 MB) to each embryo development entry.	2	Supports UC_LT_02.
FR-LT-09	The system shall allow the lab technician to record cryopreservation details: storage tank, cane, straw position, freezing date, and vitrification method.	1	Supports UC_LT_02.
FR-LT-10	The system shall update the embryo status to 'Frozen' upon successful cryopreservation record entry.	1	Supports UC_LT_02.
FR-LT-11	The system shall display any clinical notes or freeze/transfer instructions added by the doctor to the embryo record.	2	Supports UC_LT_02.

FR-LT-12	The system shall allow the lab technician to add, update, and view donation bank stock (sperm, egg, embryo) including quantity, collection date, expiry date, and storage location.	1	Supports UC_LT_03.
FR-LT-13	The system shall allow the lab technician to update a donor sample's screening status (Pending, Cleared, Rejected) with a mandatory reason note.	1	Supports UC_LT_03.
FR-LT-14	The system shall automatically remove Rejected or expired samples from the assignable pool.	1	Supports UC_LT_03.
FR-LT-15	The system shall send a low-stock alert to the lab technician and hospital administrator when any sample type falls below the configured minimum threshold.	2	Supports UC_LT_03.
FR-LT-16	The system shall allow the lab technician to log chain-of-custody events (Received, Processed, Stored, Transferred, Discarded) linked to a sample by barcode or ID.	2	Supports UC_LT_04.
FR-LT-17	The system shall require a mandatory reason code for Discarded events and record the recipient identity for Transferred events.	2	Supports UC_LT_04.
FR-LT-18	The system shall store all custody events as immutable records with timestamp and technician ID.	1	Supports UC_LT_04.
FR-LT-19	The system shall allow the lab technician to view and update equipment records including instrument name, last calibration date, next calibration due date, and status.	2	Supports UC_LT_05.
FR-LT-20	The system shall highlight instruments whose next calibration date is within 7 days or overdue.	2	Supports UC_LT_05.
FR-LT-21	The system shall allow the lab technician to record QC checks (Pass/Fail) per instrument or reagent lot, with mandatory corrective action note on Fail.	1	Supports UC_LT_05.
FR-LT-22	The system shall block the use of any instrument with a failed QC or overdue calibration from patient testing until cleared.	1	Supports UC_LT_05.

FR-LT-23	The system shall allow the lab technician to schedule and record preventive maintenance tasks with date and technician responsible.	2	Supports UC_LT_05.
FR-LT-24	The system shall generate a daily lab activity summary report (tests processed, pending, flagged, equipment issues) viewable by the lab supervisor.	3	Supports US_LT_15.

3. Non-Functional Requirements

The following non-functional requirements define the quality attributes of the system as they apply to the Lab Technician's experience and the data the technician manages.

Category	Req #	Requirement	Pri.
Performance	**NFR-LT-01**	Test report upload (up to 25 MB) shall complete within 5 seconds on a standard hospital network.	1
Performance	**NFR-LT-02**	The test queue page shall load and display up to 100 pending orders within 2 seconds.	1
Reliability	**NFR-LT-03**	The system shall maintain 99.9% uptime during clinic operating hours (06:00–22:00).	1
Security	**NFR-LT-04**	All lab technician actions shall require authenticated sessions with role-based access control.	1
Security	**NFR-LT-05**	Audit logs shall be immutable and tamper-evident; all entries must include technician ID, timestamp, and the record modified.	1
Security	**NFR-LT-06**	The system shall comply with GDPR and Albanian Law No. 9887 on Protection of Personal Data.	1
Usability	**NFR-LT-07**	The test queue, embryology module, and donation bank screens shall be navigable within 3 clicks from the dashboard.	2
Usability	**NFR-LT-08**	Error messages shall appear inline next to the field within 500 ms and clearly state what is wrong.	2
Usability	**NFR-LT-09**	The interface shall follow a consistent sidebar navigation pattern with clearly labelled sections.	1

Scalability	**NFR-LT-10**	The system shall support up to 500 concurrent lab technician sessions without degradation.	2
Maintainability	**NFR-LT-11**	Clinical reference ranges and QC thresholds shall be configurable by the lab manager without code deployment.	1
Compliance	**NFR-LT-12**	All QC records and calibration logs shall be retained per ISO 15189 and local regulatory authority.	1
Interoperability	**NFR-LT-13**	The API shall expose RESTful endpoints accepting and returning JSON for third-party LIS integration.	1
Traceability	**NFR-LT-14**	All embryo development entries shall be append-only; corrections must be new entries referencing the original.	1
Availability	**NFR-LT-15**	Urgent abnormal-result alerts shall be delivered to the doctor within 30 seconds of submission.	1
Portability	**NFR-LT-16**	The interface shall be a responsive web application compatible with Chrome, Firefox, Safari, and Edge.	1
Portability	**NFR-LT-17**	The interface shall adapt to screen sizes from 768px (tablet) to 1920px (desktop) without loss of functionality.	2
Data Protection	**NFR-LT-18**	Patient biological sample records and donor data shall be retained for a minimum of 10 years.	1

4. Lab Technician User Stories

This section contains 15 user stories for the Lab Technician role: 3 refined from the original requirements and 12 newly identified during analysis. Each story includes acceptance criteria that define when the story is considered complete.

Priorities: High = must-have for MVP (Sprint 1–2); Medium = important for Sprint 3–4; Low = nice-to-have.

ID	User Story	Acceptance Criteria	Pri.	Status
US_LT_01	As a lab technician, I want to view the queue of pending lab test requests ordered by urgency and due time so that I can prioritise and process the most critical samples first.	Queue displays all pending orders sorted by urgency then due time. Urgent orders are visually highlighted. Queue refreshes automatically or on demand.	High	New

US_LT_02	As a lab technician, I want to receive and label patient samples upon arrival by scanning a barcode or entering a sample ID.	Sample is registered with a unique ID via barcode scan or manual entry. System creates a 'Received' custody event automatically. Timestamp and technician ID are recorded.	High	New
US_LT_03	As a lab technician, I want to upload completed test reports linked to the patient record and the requesting doctor's order.	Report is linked to both patient record and doctor's order. Accepted formats: PDF, JPEG, PNG, CSV (max 25 MB). Doctor is notified that results are ready.	High	Refined
US_LT_04	As a lab technician, I want the system to automatically flag abnormal test results based on predefined clinical reference ranges and send an urgent alert to the requesting doctor.	System compares values against predefined reference ranges. Abnormal values are flagged with a colour-coded indicator. Urgent alert sent to doctor within 30 seconds.	High	Refined
US_LT_05	As a lab technician, I want to record daily embryo development observations — including cell count, fragmentation %, morphology score, and free-text notes.	Mandatory fields: development day, cell count, morphology grade. Values outside expected ranges trigger a warning. Entry is saved as immutable with timestamp and technician ID.	High	Refined
US_LT_06	As a lab technician, I want to record embryo cryopreservation details (storage tank, cane, straw, date, vitrification method).	All cryopreservation fields are mandatory. Embryo status updates to 'Frozen' upon successful entry. Storage location is linked to the embryo record.	High	New
US_LT_07	As a lab technician, I want to view any clinical notes or transfer/freeze instructions added by the doctor to an embryo record.	Doctor instructions displayed prominently on the embryo record. Instructions are read-only for the technician. New instructions trigger a visual notification.	Medium	New

US_LT_08	As a lab technician, I want to update donation bank stock levels (sperm, egg, and embryo quantities, collection dates, and storage locations).	Technician can add new samples with donor ID, type, quantity, date, and location. System validates donor ID against registered donors. A 'Received' custody event is logged automatically.	High	Refined
US_LT_09	As a lab technician, I want to update a donor's sample screening status (Pending, Cleared, Rejected).	Status change requires a mandatory reason note. Rejected samples are immediately removed from the assignable pool. All status changes logged with technician ID and timestamp.	High	New
US_LT_10	As a lab technician, I want to log every sample chain-of-custody event with a timestamp and my technician ID.	Supported event types: Received, Processed, Stored, Transferred, Discarded. Transferred events require recipient identity. Discarded events require mandatory reason code. All entries are immutable.	Medium	Refined
US_LT_11	As a lab technician, I want to view and update lab equipment records (instrument name, calibration dates, maintenance status).	Equipment list displays name, last calibration, next due, and status. Overdue instruments are highlighted and blocked from patient testing. Updates saved with timestamp and technician ID.	Medium	Refined
US_LT_12	As a lab technician, I want to schedule and record preventive maintenance tasks for lab equipment.	Maintenance task includes date, technician responsible, and notes. Completed tasks update the equipment's maintenance history.	Medium	New
US_LT_13	As a lab technician, I want to perform and record QC checks for reagents and instruments (pass/fail) with corrective action.	Fail result requires mandatory corrective action note. Failed instruments blocked from patient testing. Pass clears the block. All records include timestamp and technician ID.	High	New

US_LT_14	As a lab technician, I want to receive a low-stock alert when critical lab supplies fall below a configurable minimum threshold.	Alert sent to lab technician and hospital administrator. Threshold is configurable by lab manager. Samples expiring within 30 days are flagged.	Medium	New
US_LT_15	As a lab technician, I want to generate a daily lab activity summary report.	Report shows: tests processed, pending, flagged, equipment issues. Report is viewable by the lab supervisor. Report can be exported or printed.	Low	New

5. User Scenarios List

The following table summarises all user scenarios for the Lab Technician role. Each scenario is expanded in detail in Section 6.

ID	Scenario Name	Description
SC_LT_01	Process urgent test result	Technician opens the test queue, processes an urgent sample, uploads the report, and the system auto-flags a critically low value and alerts the doctor.
SC_LT_02	Record day-3 embryo observations	Technician opens the embryology module, enters day-3 data (cell count, fragmentation, grade), optionally attaches a microscopy image.
SC_LT_03	Freeze an embryo for FET	Technician selects 'Freeze Embryo', enters storage location and vitrification details; embryo status updates to Frozen.
SC_LT_04	Receive a donated sperm sample	Technician scans barcode, enters donor ID and sample details; system logs a 'Received' custody event with status Pending.
SC_LT_05	Clear a donor sample after screening	Technician updates donor sample status from Pending to Cleared with a mandatory reason note.
SC_LT_06	Track a sample through processing	Technician scans sample, views custody log, selects 'Processed'; system records the event with timestamp and technician ID.
SC_LT_07	Log a failed QC check	During morning QC, the analyser fails; technician records QC: Fail with corrective action; instrument is blocked from patient testing.
SC_LT_08	Schedule calibration for overdue instrument	Technician notices overdue calibration, updates the calibration date and status; system clears the overdue flag.
SC_LT_09	Receive a low-stock alert	Cryoprotectant reagent drops below minimum; system alerts technician and administrator.

SC_LT_10	Generate daily lab summary	At end of shift, technician generates summary: tests processed, pending, flagged, equipment issues.

6. User Scenarios Extended

Each user scenario is expanded below with step-by-step flows, including both the normal path and exception handling.

SC_LT_01: Process urgent test result

1. Lab Technician opens the test queue from the dashboard.
2. System displays pending orders sorted by urgency; urgent order appears at the top.
3. Technician selects the urgent order; system displays patient name, test type, requesting doctor, and clinical notes.
4. Technician processes the sample and uploads the result PDF.
5. System validates the file (format and size ≤ 25 MB).
6. System compares result values against predefined clinical reference ranges.
7. FSH value is critically low — system auto-flags with colour indicator and sends urgent alert to the requesting doctor.
8. System saves the report linked to the patient record and doctor order.
9. Audit log records: technician ID, timestamp, order reference.
10. If file exceeds 25 MB: system rejects with an inline error; technician corrects and re-uploads.
11. If doctor order is not found: technician searches by patient ID or doctor name.

SC_LT_02: Record day-3 embryo observations

1. Technician navigates to the Embryology Module.
2. Technician searches for and selects Arta's active IVF cycle.
3. System displays the embryo list with latest data and any doctor instructions.
4. Technician selects the embryo to update.
5. Technician enters day-3 data: 8 cells, 10% fragmentation, grade BB, free-text note.
6. Technician optionally attaches a microscopy image (max 10 MB).
7. System validates mandatory fields (day, cell count, grade).
8. System saves the entry as immutable with timestamp and technician ID.
9. Doctor and patient dashboards update in near-real-time.
10. If a mandatory field is missing: system blocks submission and highlights the field.
11. If image exceeds 10 MB: system rejects with an error.

SC_LT_03: Freeze an embryo for FET

1. Technician opens the embryo record in the Embryology Module.
2. System displays the doctor's freeze instruction prominently.
3. Technician selects 'Freeze Embryo'.
4. Technician enters: tank C4, cane 2, straw 3, vitrification method, today's date.
5. System validates all cryopreservation fields are complete.

6. System saves the cryopreservation record and updates embryo status to Frozen.
7. Storage location is linked to the embryo record; audit log is updated.
8. If storage location fields are incomplete: system blocks submission.
9. If cycle has been cancelled by doctor: technician can add a final note but cannot freeze.

SC_LT_04: Receive a donated sperm sample

1. Technician navigates to the Donation Bank module.
2. Technician selects 'Add New Sample'.
3. Technician scans barcode or enters donor ID D-055, sample type (sperm), quantity, collection date, storage location.
4. System validates donor ID against registered donor records.
5. System adds the sample with status Pending and logs a 'Received' custody event.
6. If donor ID is not found: system prompts to register the donor first.

SC_LT_05: Clear a donor sample after screening

1. Technician opens the Donation Bank module and finds donor D-042.
2. Technician selects 'Update Screening Status'.
3. Technician changes status from Pending to Cleared.
4. Technician enters a mandatory reason note.
5. System saves the change; sample becomes eligible for patient assignment.
6. If mandatory reason note is missing: system blocks submission.
7. If status is set to Rejected: sample is immediately removed from the assignable pool.

SC_LT_06: Track a sample through processing

1. Technician navigates to the Sample Tracking module.
2. Technician scans barcode for sample LB-2091.
3. System retrieves and displays the current custody history (Received → Stored).
4. Technician selects event type: 'Processed' and confirms.
5. System records the event with timestamp and technician ID.
6. Full chain-of-custody history is available for audit.
7. If sample ID is not found: system returns an error and prompts to verify or register the sample.

SC_LT_07: Log a failed QC check

1. Technician navigates to Equipment Management module.
2. Technician selects the hormone analyser.
3. Technician selects 'Record QC Check'.
4. Technician enters QC result values and selects Fail.
5. Technician enters corrective action: 'Reagent lot R-22 replaced'.
6. System saves QC record and flags instrument as 'Pending QC clearance'.
7. System blocks instrument from patient testing until a Pass QC is entered.
8. If corrective action field is left empty on a Fail result: submission is blocked.

SC_LT_08: Schedule calibration for overdue instrument

1. Technician navigates to Equipment Management module.
2. Technician selects the centrifuge (status: Calibration Required).
3. Technician updates calibration date and sets maintenance status to Active.
4. Technician adds a note documenting the calibration.
5. System saves the update with timestamp and technician ID; overdue flag is cleared.
6. Instrument is now available for patient testing.
7. If instrument has a failed QC in addition to overdue calibration: both must be resolved before release.

SC_LT_09: Receive a low-stock alert

1. Cryoprotectant reagent drops below the configured minimum threshold.
2. System sends a low-stock alert to the lab technician and hospital administrator.
3. Alert includes sample type, current count, and threshold value.
4. Technician reviews the alert and initiates a restocking request.
5. Technician confirms restocking with the supplier or administrator.

SC_LT_10: Generate daily lab summary

1. At end of shift, technician navigates to the Reports section.
2. Technician selects 'Daily Lab Activity Summary'.
3. System generates the report: 12 tests processed, 3 pending, 1 result flagged, 1 equipment issue.
4. Summary is displayed on screen and available for the lab supervisor.
5. Technician can export or print the report.

7. Use Cases – Lab Technician User Flows

Each use case describes the full user flow for the lab technician, following the structured format from the course case study. Each includes: description, preconditions, main flow, alternative/exception paths, what can go wrong, postconditions, restrictions and constraints, related user stories, and dependencies.

7.1 UC_LT_01: Upload Test Report

UC Name	Upload Test Report
UC ID	UC_LT_01
Actors	Lab Technician (primary), Doctor (notified), Patient (indirect via portal)
Description	The lab technician processes a pending lab test order, uploads or enters the completed test results, and the system links the report to the patient record and the doctor's order. Abnormal values are automatically flagged and trigger an urgent alert to the requesting doctor.
Preconditions	• Lab Technician is authenticated and logged in. • A lab test order exists in the system (created by a doctor). • The test has been performed and results are ready to upload.
Main Flow (Normal)	1. Lab Technician navigates to the Test Queue and views pending orders sorted by urgency. 2. Lab Technician selects the relevant test order. 3. System displays order details: patient name, test type, requesting doctor, and clinical notes. 4. Lab Technician uploads the result file (PDF/CSV/image) or enters result values manually. 5. Lab Technician reviews the result before submission. 6. System compares result values against predefined clinical reference ranges. 7. System saves the report linked to the patient record and the doctor's order. 8. System timestamps the submission and records the technician's ID. 9. System notifies the requesting doctor that results are ready. 10. Doctor reviews and approves visibility; patient can then view results in their portal.
Alternative / Exception Flow	4a. File invalid (wrong format or > 25 MB): System rejects with an inline error; technician corrects. 6a. Abnormal result detected: System auto-flags with colour indicator and sends urgent alert. 2a. Order not found: Technician searches by patient ID or doctor name.
What Can Go Wrong	• Technician may upload results to the wrong patient order. Mitigation: system displays patient name and test type for confirmation. • Network failure during file upload could lose the result. Mitigation: upload progress indicator and retry mechanism. • Reference range configuration may be outdated. Mitigation: lab manager reviews quarterly.
Postconditions	• Test report is stored and linked to the patient record. • Requesting doctor receives a notification. • If flagged, an urgent alert is sent immediately. • Audit log records: technician ID, timestamp, order reference.
Restrictions & Constraints	• Only authenticated lab technicians can upload results. • Results must be linked to an existing doctor-created order. • Uploaded files must not exceed 25 MB; accepted formats: PDF, JPEG, PNG, CSV. • Results cannot be deleted once submitted; corrections require an amended report with a reason. • Patient access to results is gated by doctor review.
Related User Stories	US_LT_01, US_LT_02, US_LT_03, US_LT_04

7.2 UC_LT_02: Record Embryo Development & Cryopreservation

UC Name	Record Embryo Development & Cryopreservation
UC ID	UC_LT_02
Actors	Lab Technician (primary), Doctor (views and adds instructions), Patient (read-only view)
Description	The lab technician records daily embryo development observations during the IVF culture period and, when instructed by the doctor, documents cryopreservation details for frozen embryo transfer planning.
Preconditions	• Lab Technician is authenticated and logged in. • An active IVF cycle exists with fertilised embryos. • The lab technician is assigned to this patient's cycle.
Main Flow (Normal)	1. Lab Technician navigates to the Embryology Module. 2. Lab Technician searches for and selects the patient's active IVF cycle. 3. System displays the embryo list with most recent data and any doctor instructions. 4. Lab Technician selects an embryo to update. 5. Lab Technician enters observation data: development day, cell count, fragmentation %, morphology grade, and free-text notes. 6. Lab Technician optionally uploads a microscopy image (max 10 MB). 7. System validates mandatory fields (day, cell count, grade). 8. System saves the entry as immutable, with a timestamp and technician ID. 9. System updates the embryo's status on the doctor's and patient's dashboards. 10. For cryopreservation: Lab Technician selects 'Freeze Embryo', enters storage location, freezing date, and vitrification method. 11. System records the cryopreservation event and updates embryo status to Frozen.
Alternative / Exception Flow	3a. No embryos recorded yet: System prompts for day-0 fertilisation data. 5a. Value outside expected range: System warns but allows entry with mandatory confirmation note. 7a. Mandatory field missing: System blocks submission. 6a. Cycle cancelled by doctor: Technician can add a final note but cannot transition the cycle.
What Can Go Wrong	• Technician may record data for the wrong embryo. Mitigation: system displays embryo identifier and last observation for confirmation. • Incorrect cryopreservation location could make retrieval impossible. Mitigation: all storage fields are mandatory and validated.

Postconditions	• Embryo record is updated; entry is immutable once confirmed. • Doctor and patient views reflect the new data in near-real-time. • If cryopreserved, storage location is recorded and embryo status = Frozen.
Restrictions & Constraints	• Only assigned lab technicians can record embryo data. • Entries are append-only; corrections must be a new entry referencing the original. • Minimum mandatory fields: development day, cell count, morphology grade. • Cryopreservation records must include tank, cane, straw, date, and method.
Related User Stories	US_LT_05, US_LT_06, US_LT_07
Dependencies	UC_LT_01 (Upload Test Report) — fertilisation results are typically reported before embryo culture begins.

7.3 UC_LT_03: Manage Donation Bank Stock & Donor Sample Status

UC Name	Manage Donation Bank Stock & Donor Sample Status
UC ID	UC_LT_03
Actors	Lab Technician (primary), Hospital Administrator (receives low-stock alerts)
Description	The lab technician manages the clinic's donation bank inventory — adding new donor samples, updating screening statuses, and ensuring expired or rejected samples are removed from the assignable pool.
Preconditions	• Lab Technician is authenticated and logged in. • Donation Bank module is configured with at least one donor record.
Main Flow (Normal)	1. Lab Technician navigates to the Donation Bank module. 2. System displays current inventory: type, donor ID, quantity, collection date, expiry, storage location, screening status. 3. Lab Technician selects action: Add New Sample, Update Quantity, Update Screening Status, or Mark as Expired. 4. Add New Sample: Technician enters donor ID, type, quantity, date, location, status (Pending). 5. System validates donor ID against registered donors. 6. System adds sample and logs a 'Received' custody event. 7. Update Screening Status: Technician sets Cleared or Rejected with mandatory reason note. 8. System updates sample; Rejected samples removed from assignable pool. 9. System sends low-stock alert if threshold breached.
Alternative / Exception Flow	4a. Donor ID not found: System prompts to register donor first. 9a. Quantity below threshold: System triggers low-stock notification. Expiry path: System flags samples expiring within 30 days; expired samples locked.
What Can Go Wrong	• Technician may enter incorrect donor ID. Mitigation: system displays donor name for confirmation. • Stock levels may desync with physical inventory. Mitigation: periodic audit reconciliation.
Postconditions	• Inventory is updated and reflects physical stock. • All changes logged as custody events with technician ID and timestamp. • Rejected/expired samples removed from assignable pool.
Restrictions & Constraints	• All donor samples must be linked to a registered donor. • Expired or rejected samples cannot be assigned to patients. • Every inventory change requires authentication and is permanently logged.
Related User Stories	US_LT_08, US_LT_09, US_LT_14
Dependencies	None — this is an entry-point use case.

7.4 UC_LT_04: Log Sample Chain-of-Custody Event

UC Name	Log Sample Chain-of-Custody Event
UC ID	UC_LT_04
Actors	Lab Technician (primary)
Description	The lab technician logs a chain-of-custody event for a patient or donor sample, maintaining full traceability from receipt through processing, storage, transfer, or disposal.
Preconditions	• Lab Technician is authenticated and logged in. • A patient sample or donor sample record exists in the system.
Main Flow (Normal)	1. Lab Technician navigates to the Sample Tracking module. 2. Lab Technician scans the sample barcode or manually enters the sample ID. 3. System retrieves and displays the sample's current custody history. 4. Lab Technician selects event type: Received, Processed, Stored, Transferred, or Discarded. 5. For 'Transferred': Technician enters the destination. 6. For 'Discarded': Technician enters disposal reason code and method. 7. System records the event with: event type, timestamp, technician ID, and notes. 8. System displays the updated chain-of-custody log.
Alternative / Exception Flow	2a. Sample ID not found: System returns error; technician must verify or register. 4a. Discarded event: System requires mandatory reason code and disposal confirmation.
What Can Go Wrong	• Technician may scan the wrong barcode. Mitigation: system displays sample details for confirmation. • Disposal may be logged without proper authorisation. Mitigation: controlled reason code list and confirmation step.
Postconditions	• A new chain-of-custody entry is created and linked to the sample. • The entry is immutable. • Full custody history is available for regulatory audit.
Restrictions & Constraints	• All custody events must be linked to an authenticated technician. • Event records are immutable once created. • Disposal events require a mandatory reason code. • Transfer events must record recipient identity.
Related User Stories	US_LT_10
Dependencies	UC_LT_01 or UC_LT_03 — a sample must exist in the system.

7.5 UC_LT_05: Manage Lab Equipment Records & Perform QC Checks

UC Name	Manage Lab Equipment Records & Perform QC Checks
UC ID	UC_LT_05
Actors	Lab Technician (primary)
Description	The lab technician views and updates equipment calibration records, performs and documents QC checks, and schedules preventive maintenance to ensure all instruments meet clinical compliance standards.
Preconditions	• Lab Technician is authenticated and logged in. • Equipment records are registered in the Equipment Management module. • QC reference ranges and pass/fail criteria are configured.
Main Flow (Normal)	1. Lab Technician navigates to the Equipment Management module. 2. System displays equipment list: name, last calibration, next due, and status. 3. System highlights instruments overdue or due within 7 days. 4. Lab Technician selects an instrument to update. 5. Lab Technician updates calibration date or maintenance status and adds a note. 6. System saves the update with timestamp and technician ID. 7. For QC check: Lab Technician selects 'Record QC Check'. 8. Lab Technician enters QC result values and selects Pass or Fail. 9. If Fail: Lab Technician enters the corrective action taken. 10. System saves the QC record; failed instruments are blocked from patient testing. 11. If Pass on previously failed instrument: system clears the block.
Alternative / Exception Flow	6a. Calibration overdue: System sets status to 'Calibration Required'; instrument blocked. 8a. QC Fail: System blocks instrument or reagent lot until Pass is entered. 3a. Equipment not found: Technician can create a new record.
What Can Go Wrong	• Technician may forget to record a QC failure. Mitigation: system requires QC check before instrument can process patient samples. • Calibration may be recorded but not performed. Mitigation: certificate attachment (future enhancement).
Postconditions	• Equipment record is updated with latest calibration, status, and QC results. • Instruments with failed QC or overdue calibration are flagged and blocked. • Corrective actions are documented for audit.

Restrictions & Constraints	• Only authenticated lab technicians can update equipment or QC records. • Overdue or failed-QC instruments cannot be used for patient testing. • QC checks must reference a specific reagent lot or instrument serial number. • All records are immutable once saved. • Calibration intervals and QC thresholds are set by the lab manager.
Related User Stories	US_LT_11, US_LT_12, US_LT_13
Dependencies	None — this is an entry-point use case.

8. Use Case Diagram

The Use Case Diagram below describes the Lab Technician actor and all use cases they interact with. The diagram follows UML notation with «include» and «extend» relationships where applicable.

Actor:

- Lab Technician (Silvja Fejzulla & Arbri Hamzallari)

Primary Use Cases (direct interactions):

- Upload Test Report – Technician uploads completed test results linked to the patient record and doctor order.
- View Test Queue – Technician views pending lab test orders prioritised by urgency.
- Record Embryo Development – Technician records daily embryo observations.
- Freeze Embryo (Cryopreservation) – Technician records cryopreservation details and updates embryo status to Frozen.
- View Doctor Instructions – Technician views clinical notes and freeze/transfer instructions from the doctor.
- Add Donor Sample – Technician registers a new donor sample into the donation bank inventory.
- Update Screening Status – Technician marks donor samples as Cleared or Rejected.
- Update Stock Quantity – Technician adjusts donation bank stock levels.
- Log Custody Event – Technician logs chain-of-custody events for samples.
- Update Equipment Record – Technician updates calibration dates and maintenance status.
- Record QC Check – Technician performs and records quality control checks (Pass/Fail).
- Schedule Preventive Maintenance – Technician schedules and records maintenance tasks.
- Generate Daily Lab Summary – Technician generates end-of-shift activity report.

«include» Relationships:

- Upload Test Report «include» Validate File – File format and size are always validated.
- Upload Test Report «include» Compare Reference Ranges – Result values are always compared.
- Upload Test Report «include» Send Notification – Doctor is always notified.
- Upload Test Report «include» Log Audit Trail – Every upload is recorded.
- Record Embryo Development «include» Validate Mandatory Fields – Day, cell count, and grade are always validated.
- Record Embryo Development «include» Log Audit Trail – Every entry is timestamped.
- Freeze Embryo «include» Validate Storage Fields – Tank, cane, straw, date, and method always validated.
- Add Donor Sample «include» Validate Donor ID – Donor ID always checked against registered donors.
- Add Donor Sample «include» Log Custody Event (Received) – A 'Received' event is always created.
- Update Screening Status «include» Log Audit Trail – Every status change is permanently logged.
- Log Custody Event «include» Log Audit Trail – Every custody event is recorded.
- Record QC Check «include» Log Audit Trail – Every QC result is permanently logged.

«extend» Relationships:

- Upload Test Report «extend» Flag Abnormal Result – Extends only when a result value falls outside the reference range.

- Upload Test Report «extend» Send Urgent Alert – Extends only when an abnormal result is flagged.
- Record Embryo Development «extend» Warn Out-of-Range Value – Extends only when observation is outside expected range.
- Record Embryo Development «extend» Attach Microscopy Image – Extends only when image is attached.
- Freeze Embryo «extend» Display Doctor Freeze Instruction – Extends only when doctor has added instruction.
- Update Screening Status «extend» Remove from Assignable Pool – Extends only when status is Rejected.
- Add Donor Sample «extend» Trigger Low-Stock Alert – Extends only when stock falls below minimum.
- Log Custody Event «extend» Require Disposal Confirmation – Extends only when event type is Discarded.
- Log Custody Event «extend» Record Recipient Identity – Extends only when event type is Transferred.
- Record QC Check «extend» Block Instrument – Extends only when QC result is Fail.
- Record QC Check «extend» Clear QC Block – Extends only when Pass is entered for previously failed instrument.
- Update Equipment Record «extend» Block Overdue Instrument – Extends only when calibration is overdue.

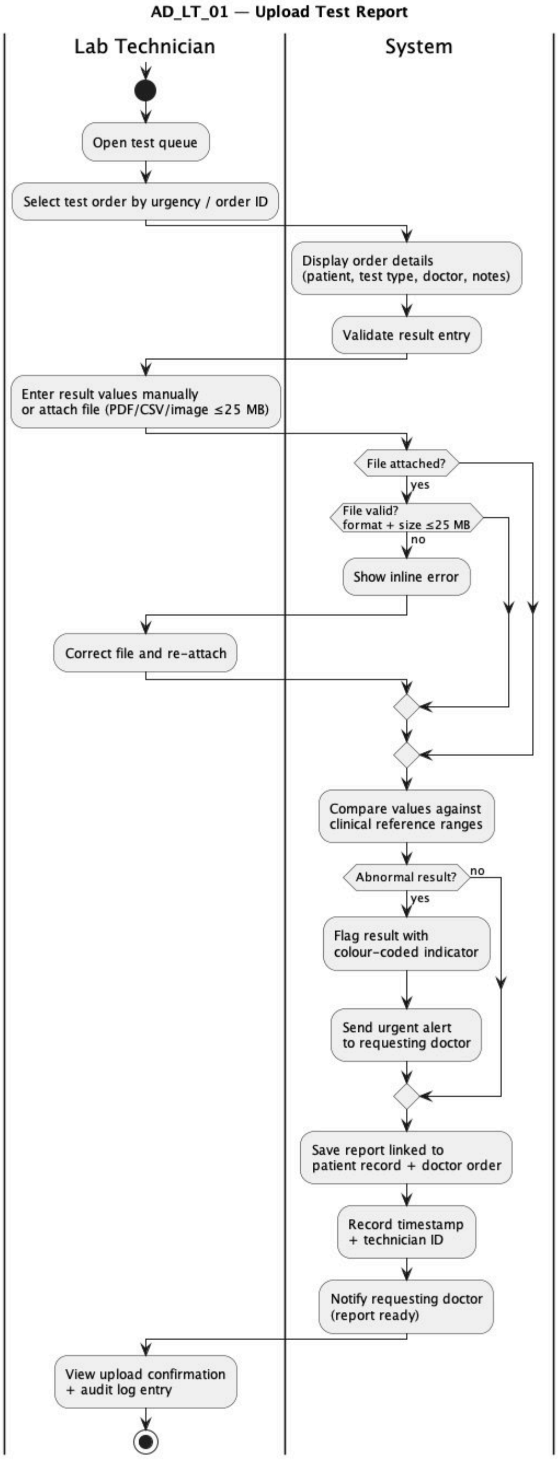
9. Activity Diagrams

This section contains the activity diagrams for each lab technician use case. Each diagram illustrates the workflow of the use case including decision points, alternative paths, and exception handling. The diagrams use UML activity diagram notation with swimlanes separating the Lab Technician actor from the System.

Each activity diagram follows a consistent structure: the Lab Technician swimlane (left) shows user-initiated actions, while the System swimlane (right) shows automated responses, validations, and data operations. Decision diamonds represent branching logic, and merge diamonds show where paths reconverge. Prerequisites are noted at the top of each diagram, listing any use cases that must be completed before the current workflow can begin.

9.1 UC_LT_01: Upload Test Report

Activity diagram for UC_LT_01: Upload Test Report

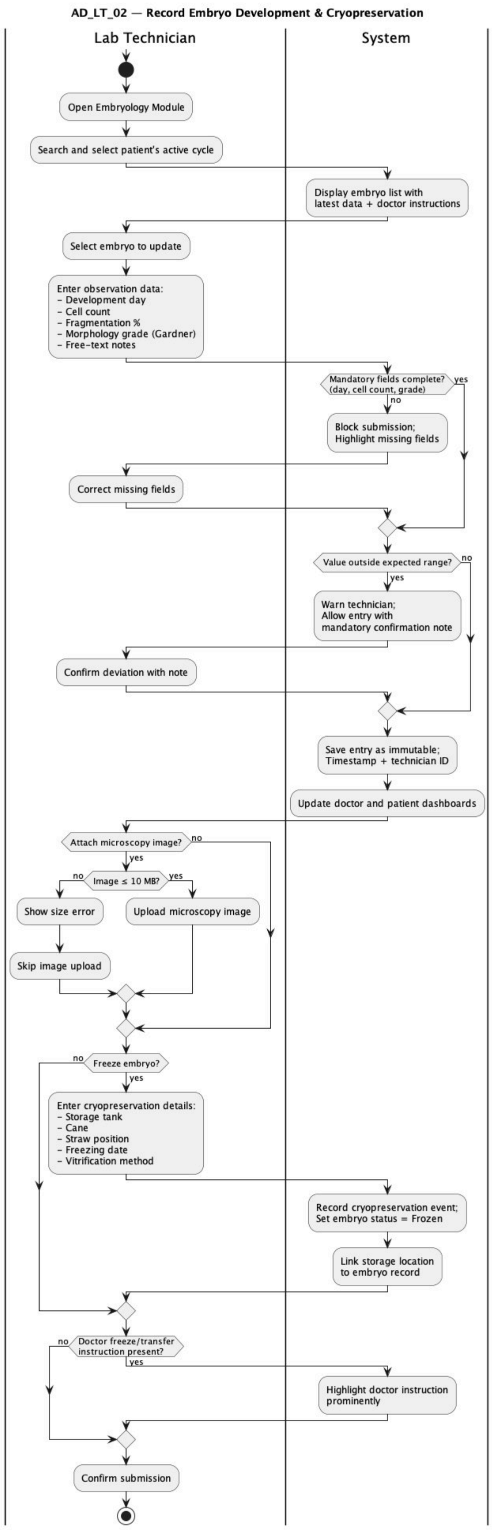


9.1 UC_LT_01: Upload Test Report — Description

This activity diagram shows the complete workflow for uploading a test report. The Lab Technician opens the test queue, selects an order by urgency, and enters result values or attaches a file. The System validates the file format and size (max 25 MB), compares values against clinical reference ranges, and flags abnormal results with a colour-coded indicator. If abnormal, an urgent alert is sent to the requesting doctor. The report is saved linked to the patient record and doctor order, with timestamp and technician ID recorded in the audit trail.

9.2 UC_LT_02: Record Embryo Development & Cryopreservation

Activity diagram for UC_LT_02: Record Embryo Development & Cryopreservation

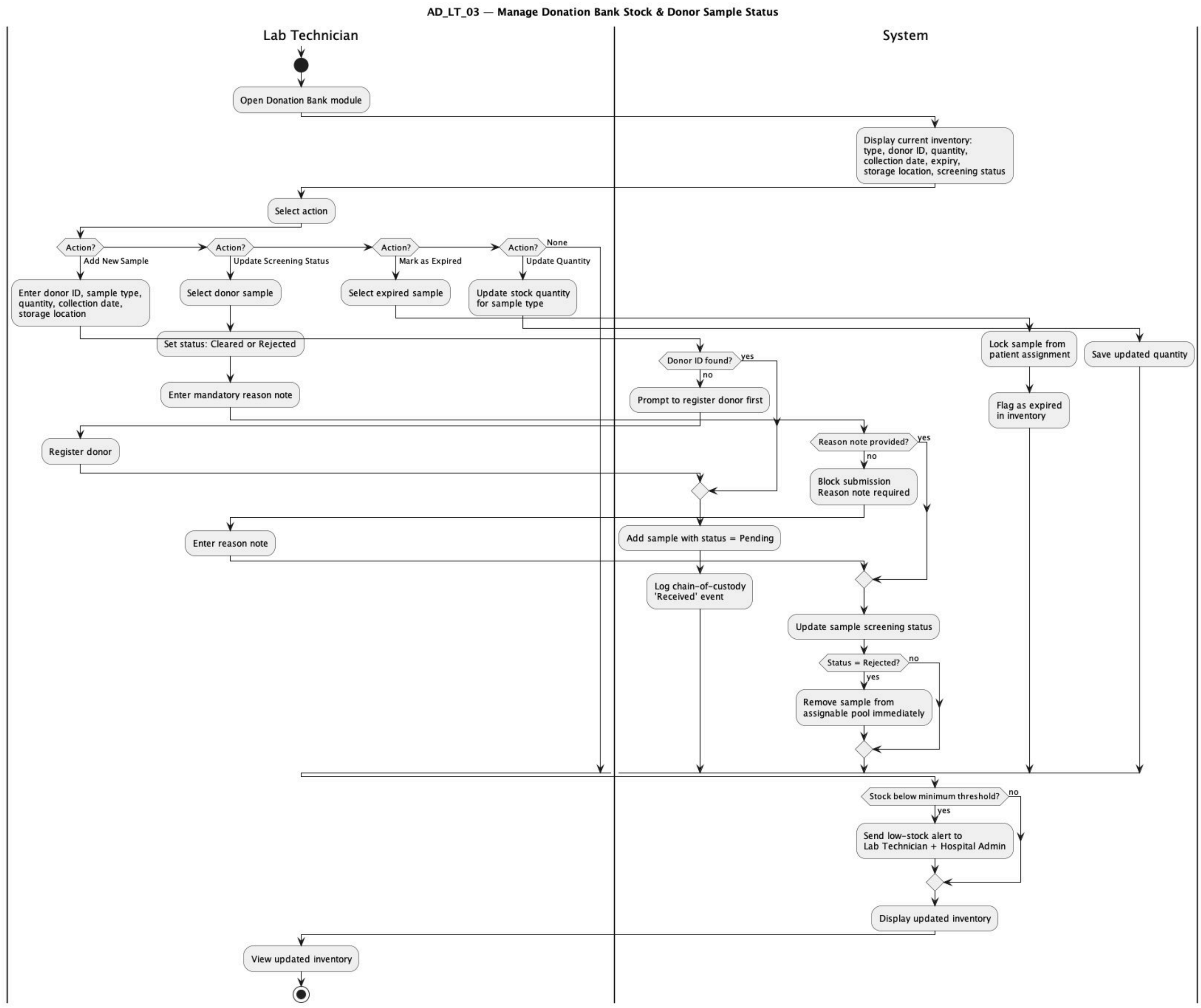


9.2 UC_LT_02: Record Embryo Development & Cryopreservation — Description

This diagram covers the daily embryo observation workflow. The technician opens the Embryology Module, selects the patient's active cycle, and enters observation data. The System validates mandatory fields and warns about out-of-range values while allowing entry with a confirmation note. Optional microscopy image upload (max 10 MB) is handled. The cryopreservation sub-flow covers freezing an embryo with storage location details, updating the embryo status to Frozen.

9.3 UC_LT_03: Manage Donation Bank Stock & Donor Sample Status

Activity diagram for UC_LT_03: Manage Donation Bank Stock & Donor Sample Status

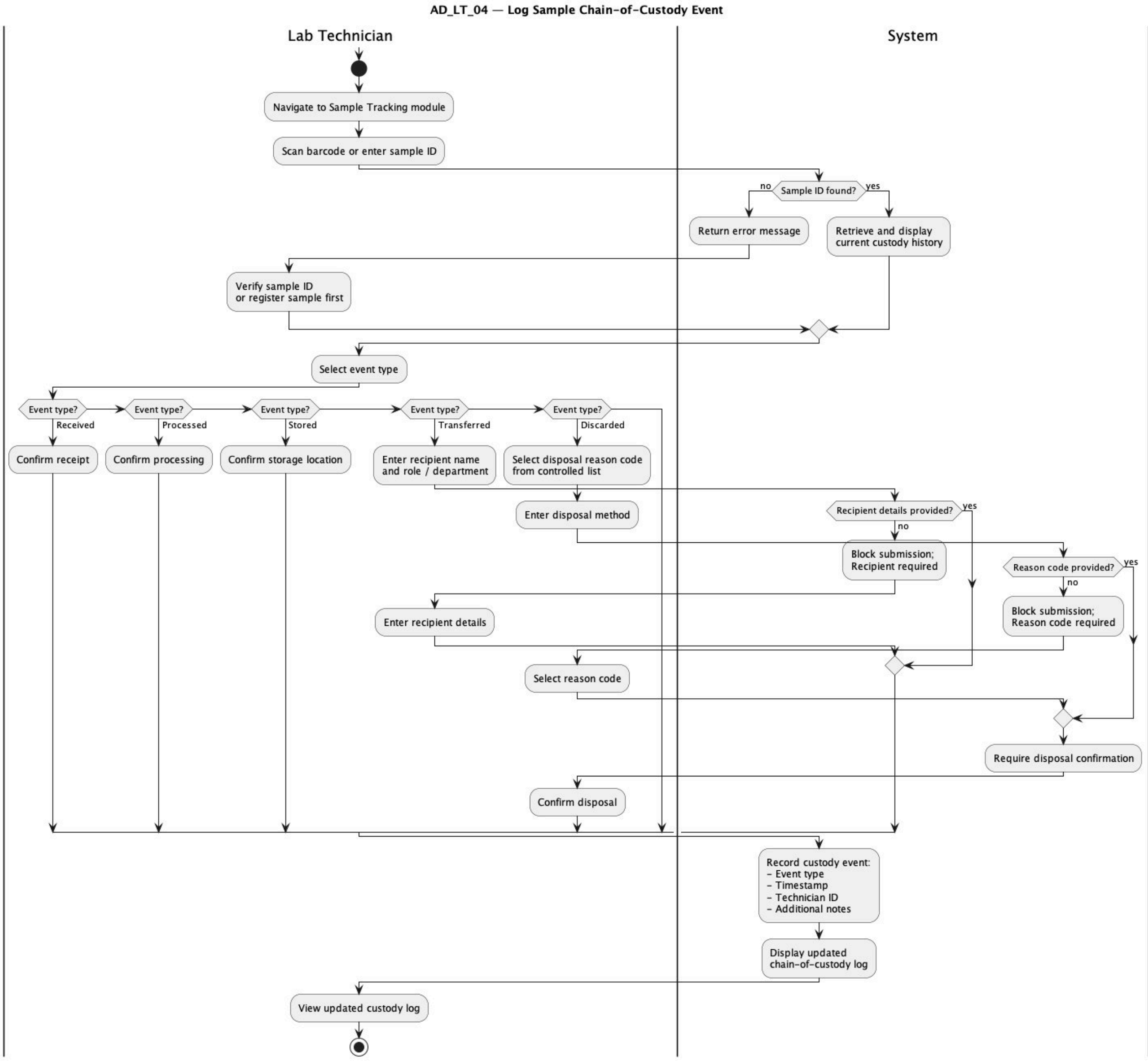


9.3 UC_LT_03: Manage Donation Bank Stock & Donor Sample Status — Description

This activity diagram illustrates four distinct action paths: Add New Sample (with donor ID validation and 'Received' custody event logging), Update Screening Status (Cleared/Rejected with mandatory reason note), Mark as Expired (locking from patient assignment), and Update Quantity. The diagram shows how rejected samples are immediately removed from the assignable pool, and how the system triggers low-stock alerts when any sample type falls below the configured minimum threshold.

9.4 UC_LT_04: Log Sample Chain-of-Custody Event

Activity diagram for UC_LT_04: Log Sample Chain-of-Custody Event

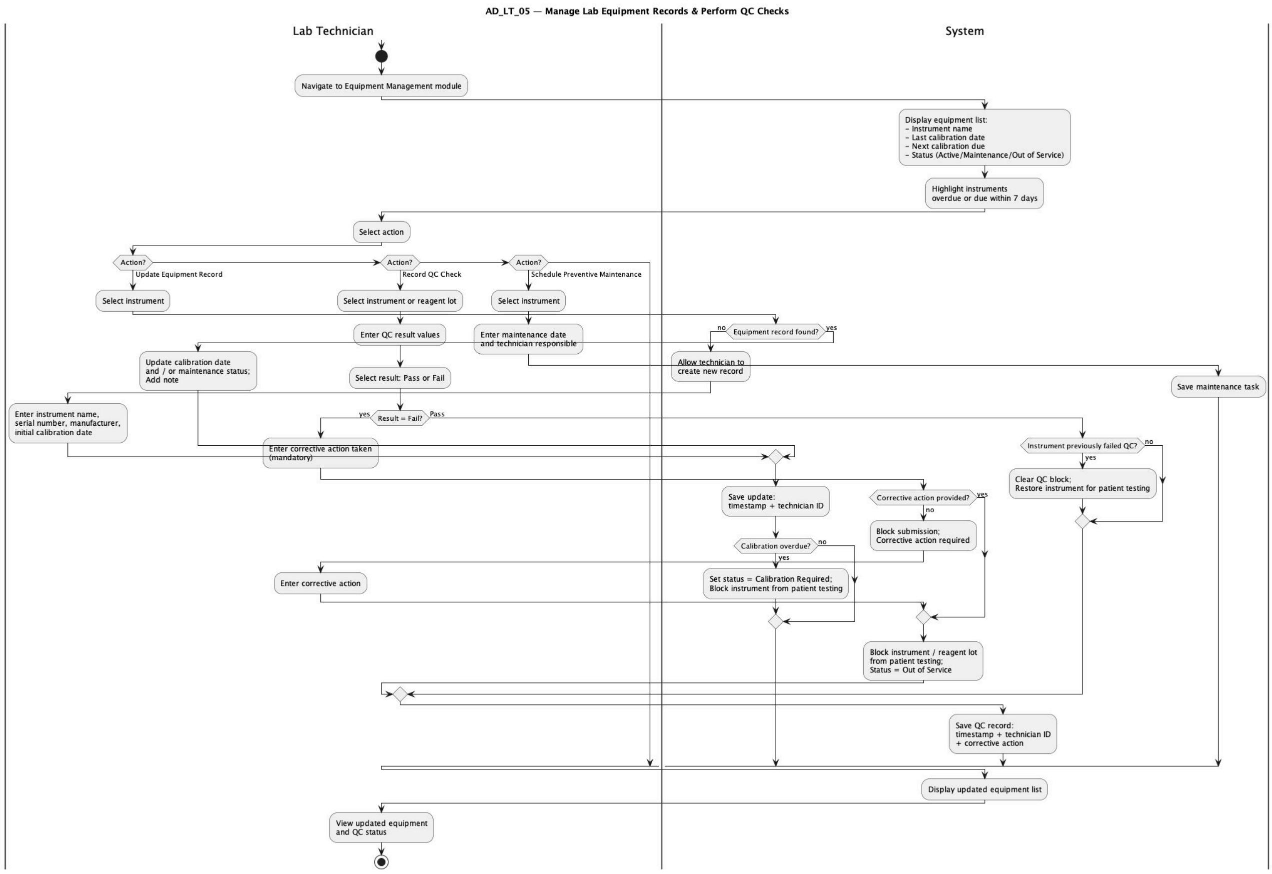


9.4 UC_LT_04: Log Sample Chain-of-Custody Event — Description

This diagram shows the custody logging workflow from barcode scan or manual sample ID entry through to the immutable event record. The System retrieves and displays the current custody history. The technician selects from five event types: Received, Processed, Stored, Transferred (requiring recipient name and role/department), and Discarded (requiring a reason code from a controlled list and disposal confirmation). All events are recorded with timestamp, technician ID, and additional notes.

9.5 UC_LT_05: Manage Lab Equipment Records & Perform QC Checks

Activity diagram for UC_LT_05: Manage Lab Equipment Records & Perform QC Checks



9.5 UC_LT_05: Manage Lab Equipment Records & Perform QC Checks — Description

This diagram covers three action paths: Update Equipment Record (calibration date, maintenance status), Record QC Check (Pass/Fail with mandatory corrective action on Fail), and Schedule Preventive Maintenance. The System highlights instruments overdue or due within 7 days. Failed QC results block the instrument from patient testing. A Pass result on a previously failed instrument clears the block and restores it for patient testing.

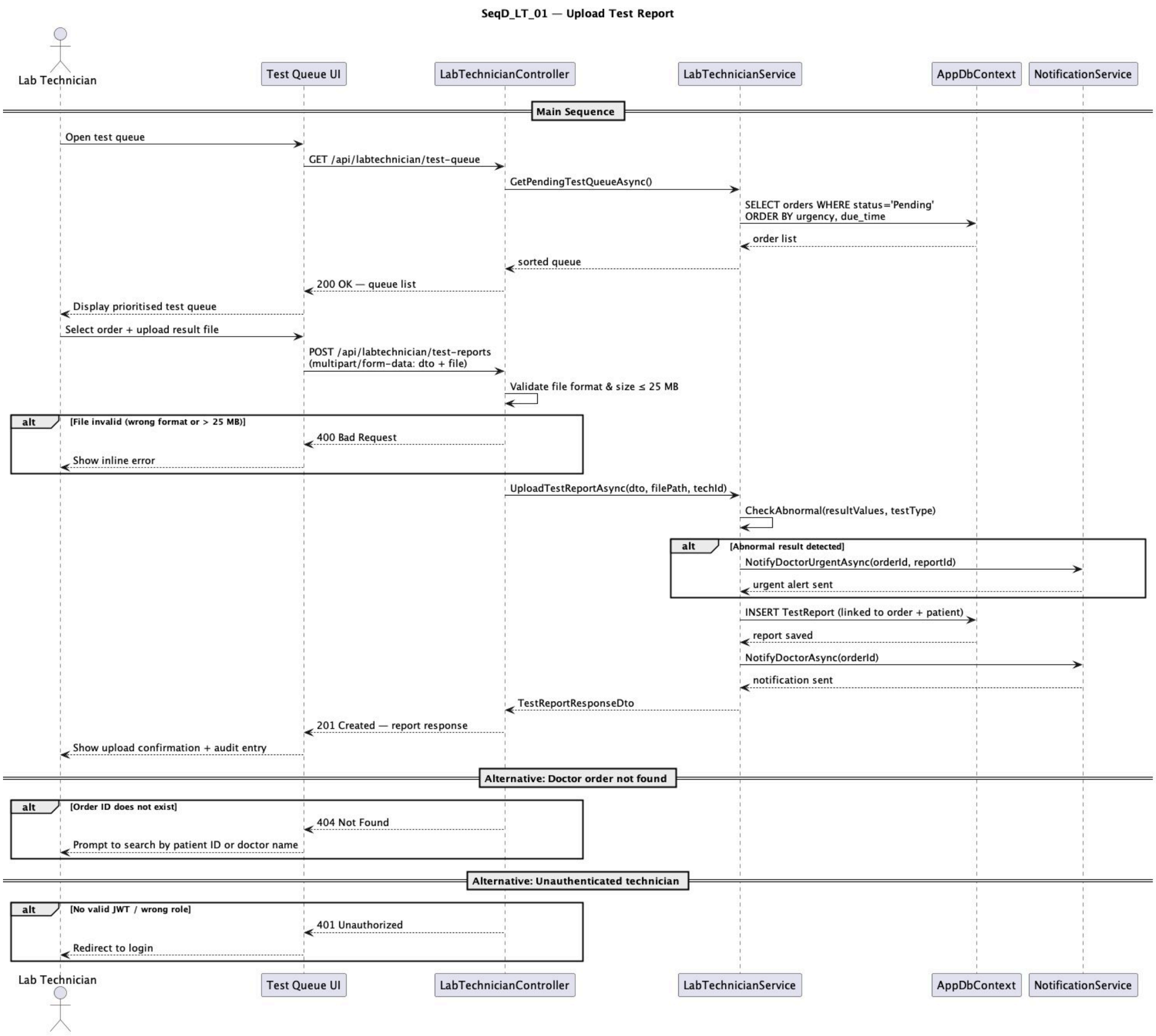
10. Sequence Diagrams

This section contains the sequence diagrams for each lab technician use case. Each diagram illustrates the chronological interactions between the lab technician, the UI layer, backend services, the database, and external components such as notification services.

Each sequence diagram follows the layered architecture of the system: the Lab Technician interacts with the UI component, which communicates with the LabTechnicianController (API Gateway). The controller delegates to the LabTechnicianService (business logic layer), which interacts with the AppDbContext (database layer) and, where needed, the NotificationService for alerts. Alternative and exception flows are shown using UML 'alt' fragments with guard conditions.

10.1 UC_LT_01: Upload Test Report

Sequence diagram for UC_LT_01: Upload Test Report

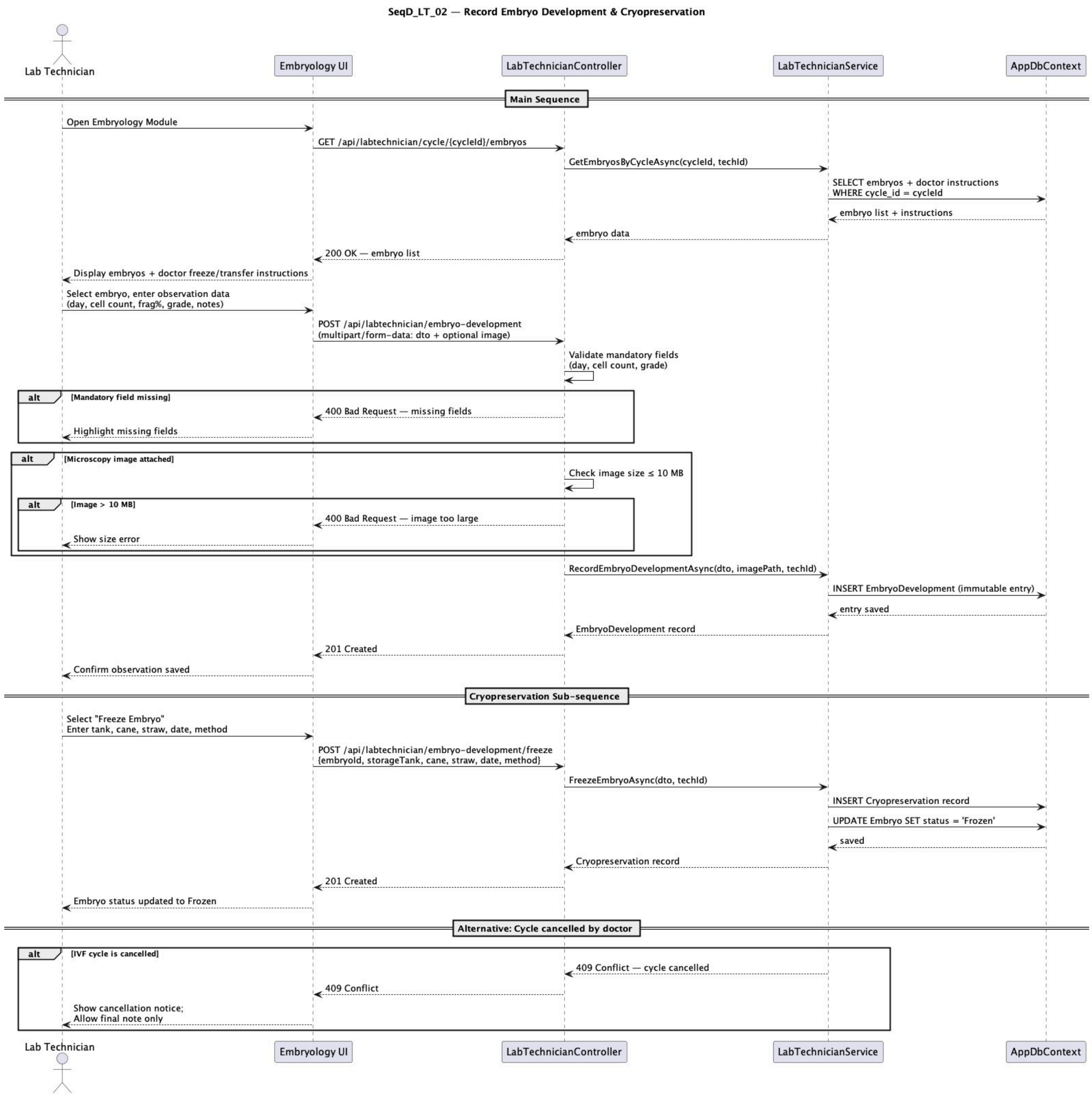


10.1 UC_LT_01: Upload Test Report — Description

Lifelines: Lab Technician, Test Queue UI, LabTechnicianController, LabTechnicianService, AppDbContext, NotificationService. The diagram shows the full request-response flow: the technician opens the test queue (GET /api/labtechnician/test-queue), the service retrieves pending orders sorted by urgency, and the UI displays the prioritised list. Upon selecting an order and uploading a result file (POST /api/labtechnician/test-reports), the controller validates file format and size. The service checks for abnormal results and dispatches an urgent alert if flagged. Alternative flows cover invalid files (400), missing orders (404), and unauthenticated access (401).

10.2 UC_LT_02: Record Embryo Development & Cryopreservation

Sequence diagram for UC_LT_02: Record Embryo Development & Cryopreservation

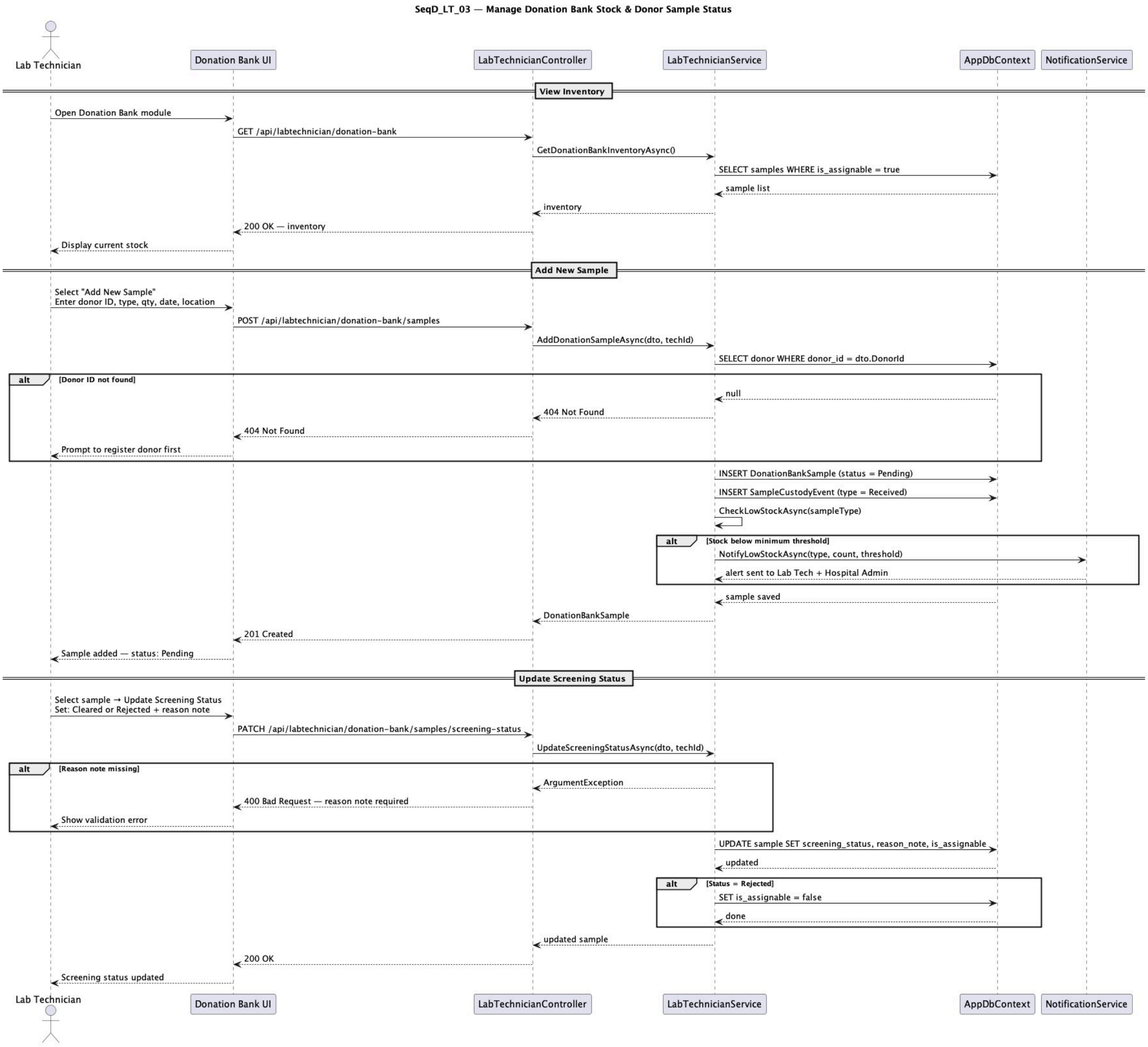


10.2 UC_LT_02: Record Embryo Development & Cryopreservation — Description

Lifelines: Lab Technician, Embryology UI, LabTechnicianController, LabTechnicianService, AppDbContext. The main sequence shows the technician opening the Embryology Module (GET /api/labtechnician/cycle/{cycleId}/embryos), the service retrieving embryos with doctor instructions, and the UI displaying them. Observation data is submitted (POST /api/labtechnician/embryo-development) with mandatory field validation. The cryopreservation sub-sequence (POST /api/labtechnician/embryo-development/freeze) inserts a Cryopreservation record and updates the embryo status to Frozen. Alternative flow handles cycle cancellation (409 Conflict).

10.3 UC_LT_03: Manage Donation Bank Stock & Donor Sample Status

Sequence diagram for UC_LT_03: Manage Donation Bank Stock & Donor Sample Status

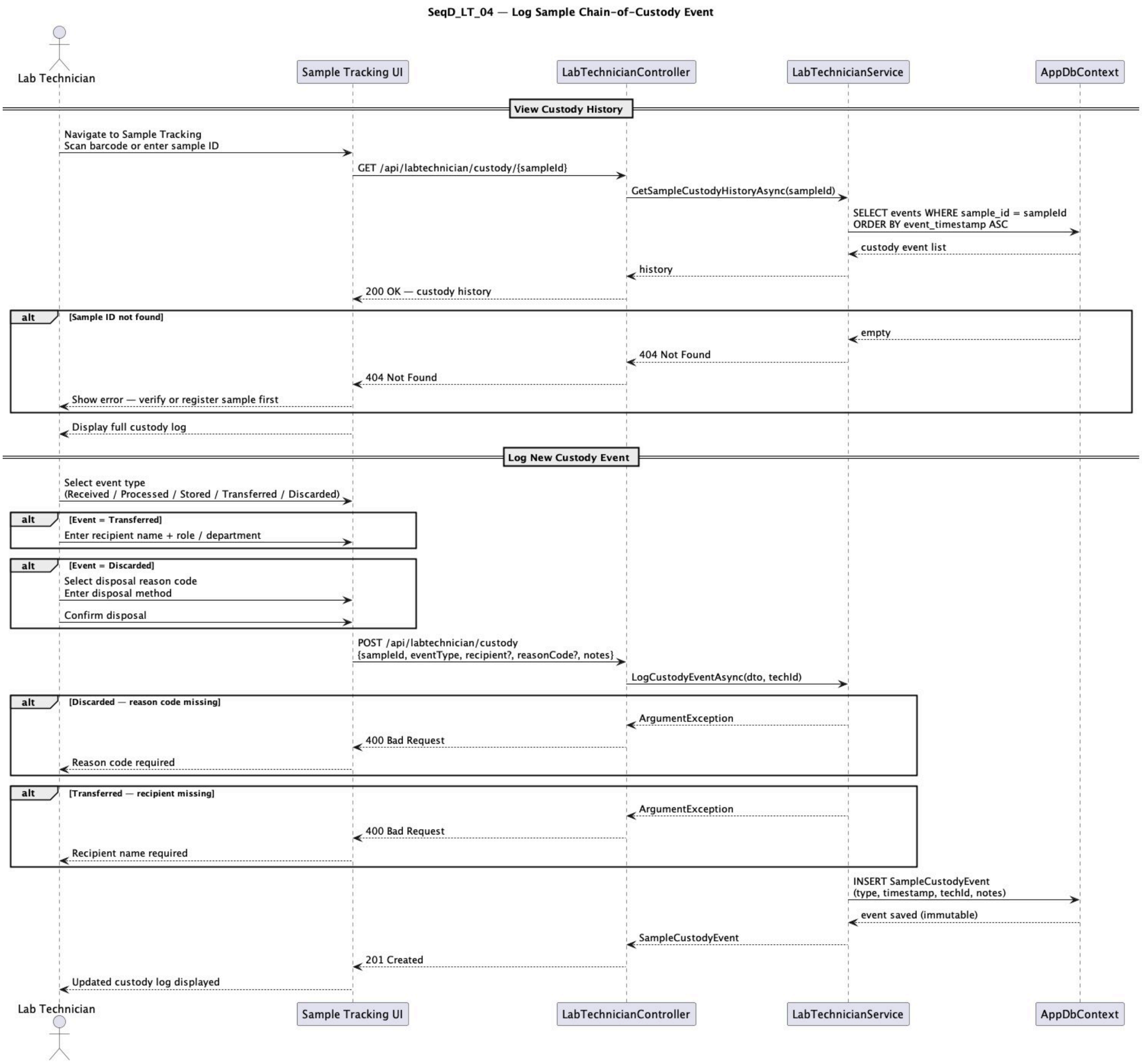


10.3 UC_LT_03: Manage Donation Bank Stock & Donor Sample Status — Description

Lifelines: Lab Technician, Donation Bank UI, LabTechnicianController, LabTechnicianService, AppDbContext, NotificationService. The diagram covers three sub-sequences: View Inventory, Add New Sample with donor ID validation and automatic 'Received' custody event logging, and Update Screening Status with mandatory reason note enforcement. The low-stock check triggers NotifyLowStockAsync when thresholds are breached. Alternative flows handle unknown donor IDs (404) and missing reason notes (400).

10.4 UC_LT_04: Log Sample Chain-of-Custody Event

Sequence diagram for UC_LT_04: Log Sample Chain-of-Custody Event



10.4 UC_LT_04: Log Sample Chain-of-Custody Event — Description

Lifelines: Lab Technician, Sample Tracking UI, LabTechnicianController, LabTechnicianService, AppDbContext. The View Custody History sub-sequence retrieves events ordered by timestamp. The Log New Custody Event sub-sequence handles all five event types. For Transferred events, the technician enters recipient name and role. For Discarded events, the technician selects a disposal reason code. The service validates mandatory fields and records the event as immutable.

10.5 UC_LT_05: Manage Lab Equipment Records & Perform QC Checks

Sequence diagram for UC_LT_05: Manage Lab Equipment Records & Perform QC Checks

10.5 UC_LT_05: Manage Lab Equipment Records & Perform QC Checks — Description

Lifelines: Lab Technician, Equipment UI, LabTechnicianController, LabTechnicianService, AppDbContext. The View Equipment List sub-sequence retrieves all instruments with overdue highlighting. The Update Equipment Record sub-sequence validates equipment existence and updates calibration dates. The Record QC Check sub-sequence enforces mandatory corrective action on Fail results, blocks failed instruments from patient testing, and restores previously blocked instruments on Pass.

11. Summary of Changes & Additions

The table below summarises all refinements made to the original Lab Technician user stories and identifies the new stories added through this analysis. The original project scope included five user stories for the Lab Technician role (US_25 through US_30). During requirements analysis, three of these were refined to improve specificity and testability, and twelve new stories were identified to cover gaps in sample tracking, equipment management, quality control, and reporting.

Story ID	Story Title	Status	Key Change / Rationale
US_LT_01	View pending test request queue	New	Provides workload visibility; prevents critical samples being missed.
US_LT_02	Receive and label patient samples on arrival	New	Closes the gap between sample collection and processing; enables full chain-of-custody from entry.
US_LT_03	Upload completed test reports (was US_25)	Refined	Added explicit linkage to doctor's order for traceability; clarified file and data entry options.
US_LT_04	Auto-flag abnormal results (was US_29)	Refined	Specified triggering mechanism (reference ranges) to make acceptance criteria testable.
US_LT_05	Record embryo development observations (was US_26)	Refined	Expanded observation fields to include fragmentation % and free-text notes.
US_LT_06	Record embryo cryopreservation details	New	Covers a critical IVF sub-process (FET planning) absent from original stories.
US_LT_07	View doctor's embryo instructions	New	Closes doctor-lab communication loop; reduces procedural errors.
US_LT_08	Update donation bank stock inventory (was US_27 part)	Refined	Split US_27 into stock quantity and donor screening for clarity.
US_LT_09	Update donor sample screening status	New	Regulatory requirement; separated from stock management for distinct traceability.
US_LT_10	Log sample chain-of-custody events (was US_30)	Refined	Added required log fields (event types, technician ID) for testable acceptance criteria.
US_LT_11	View and update equipment records (was US_28)	Refined	Made record fields explicit (calibration dates, status) for measurable acceptance criteria.
US_LT_12	Schedule and record preventive maintenance	New	Proactive maintenance scheduling to reduce unplanned downtime.
US_LT_13	Perform and record QC checks	New	ISO 15189 requirement; ensures test accuracy and reagent traceability.
US_LT_14	Receive low-stock alerts for lab supplies	New	Prevents operational disruption; threshold is configurable.
US_LT_15	Generate daily lab activity summary report	New	Supports supervisor oversight and shift handover.

12. Glossary

The following table defines key terms and abbreviations used throughout this document.

Term	Definition
IVF	In Vitro Fertilisation — a medical procedure in which an egg is fertilised by sperm outside the body, in a laboratory setting.
FET	Frozen Embryo Transfer — a procedure where a previously cryopreserved embryo is thawed and transferred to the uterus.
Cryopreservation	The process of preserving biological material (embryos, sperm, eggs) by cooling to very low temperatures using vitrification or slow-freezing methods.
Vitrification	A rapid-freezing cryopreservation technique that converts biological material to a glass-like solid state without ice crystal formation.
QC (Quality Control)	A set of procedures and tests performed on laboratory instruments and reagents to ensure accuracy, reliability, and compliance with clinical standards.
Chain-of-Custody	A documented, unbroken trail recording the sequence of custody, control, transfer, and disposition of a biological sample.
FSH	Follicle-Stimulating Hormone — a hormone measured in blood tests to assess ovarian function and fertility.
Gardner Grading	A standardised scoring system used to assess blastocyst (day-5/6 embryo) quality based on expansion, inner cell mass, and trophectoderm morphology.
Morphology Grade	A classification of embryo quality based on visual assessment of cell number, symmetry, fragmentation, and other developmental characteristics.
Fragmentation (%)	The percentage of an embryo's volume occupied by anucleate cellular fragments, used as a quality indicator.
ISO 15189	An international standard that specifies requirements for quality and competence in medical laboratories.
GDPR	General Data Protection Regulation — European Union regulation governing the processing and protection of personal data.
Albanian Law No. 9887	The Albanian law on the Protection of Personal Data, governing the collection, processing, and storage of personal information within Albania.
Audit Trail	An immutable, timestamped record of all actions performed in the system, including who performed each action, what was changed, and when.
JWT	JSON Web Token — an open standard for securely transmitting authentication information between parties.
RBAC	Role-Based Access Control — a security model that restricts system access based on the roles assigned to individual users.
LIS	Laboratory Information System — a software system used to manage laboratory data, workflows, and integration with clinical systems.
RESTful API	A web service interface that follows REST architectural constraints, using standard HTTP methods for communication.
Barcode Scan	The use of a barcode reader to identify a sample by scanning the unique barcode label attached to it.
Reagent Lot	A specific batch of chemical reagent produced under uniform conditions, identified by a unique lot number for traceability.

Donor Sample	Biological material (sperm, egg, or embryo) donated by a registered donor for use in assisted reproduction treatments.
Assignable Pool	The subset of donor samples in the donation bank that have been screened, cleared, and are eligible for assignment to patient treatments.
Blastocyst	An embryo that has developed for 5–6 days after fertilisation and has formed a fluid-filled cavity.
Oocyte	An immature egg cell (ovum) retrieved from the ovary during an IVF cycle for fertilisation in the laboratory.
Semen Analysis	A laboratory test that evaluates the quality of a sperm sample, including sperm count, motility, morphology, and volume.
Embryo Culture	The process of maintaining fertilised embryos in a controlled laboratory environment for 3–6 days.
Optimistic Locking	A concurrency control technique where the system allows multiple users to access a resource simultaneously but detects conflicts at save time.
Immutable Record	A data entry that cannot be modified or deleted once saved. Corrections must be new entries referencing the original.
Calibration	The process of comparing an instrument's measurements to a known standard to verify accuracy.
MVP	Minimum Viable Product — the initial version of the system containing only the core features required for basic operation.

End of Lab Technician Requirements Specification — Silvja Fejzulla & Arbri Hamzallari | March 2026